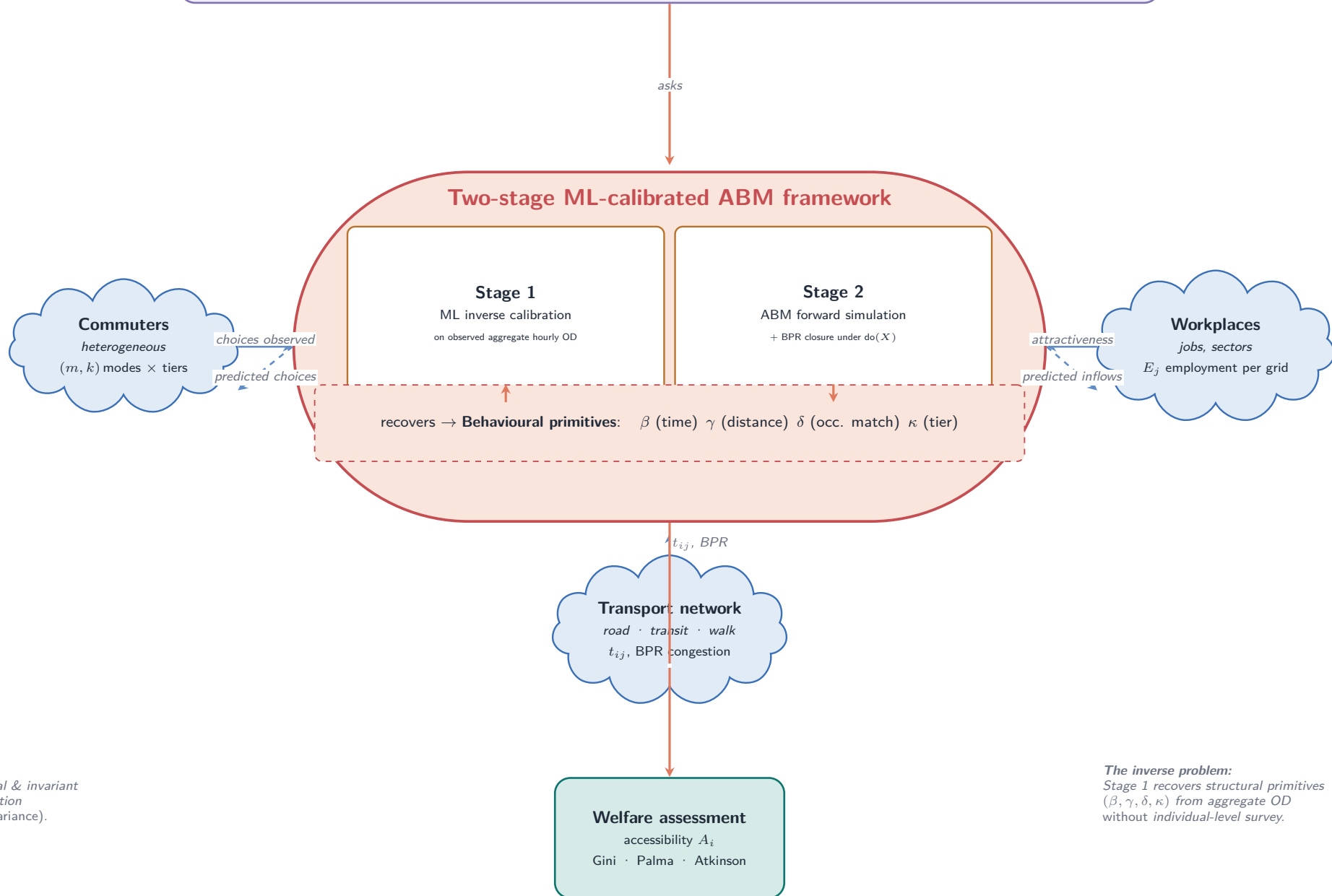


Conceptual Framework — Two-stage ML-calibrated ABM

observed commuting data → behavioural primitives → counterfactual welfare

Policy question: “what if +65,000 jobs at Old Oak Common? what if peak commute load shifted to flex-work shoulder?”



Counterfactual claim:
primitives are structural & invariant
under $do(X)$ perturbation
(single-component invariance).

The inverse problem:
Stage 1 recovers structural primitives
($\beta, \gamma, \delta, \kappa$) from aggregate OD
without individual-level survey.

Three actors — commuters (heterogeneous in mode and income), workplaces (jobs and sectors per grid), and the transport network (road/transit/walk with BPR congestion) — interact through the two-stage framework. Stage 1 calibrates behavioural primitives by inverse MLE on aggregate hourly OD; Stage 2 simulates agent-level forward dynamics with frozen primitives and BPR closure under counterfactual policy interventions. The framework's output is a welfare assessment — accessibility distribution, Gini/Palma/Atkinson inequality — for each scenario.